

and cave-ins. The soil, Sreedharan said, “became like toothpaste”; so agility with solutions and construction strategy proved to be the key. For the first time, tunnels had low-maintenance ballastless tracks, which use concrete or asphalt. Artificial ventilation, jumbo drills and electrohydraulic digging arms and loaders were imported from Sweden. Soft soil, particularly in the Pernem and Old Goa Tunnels in Goa, led to tunnel collapses, causing many delays, which were finally overcome.

The entire Konkan Railway line became operational in January 1998, changing the face of railway construction in India as engineers took their experience to other projects, such as the Delhi Metro and the Kashmir Link project (see [Kashmir Connection](#)).



▲ **The Konkan Railway** has thousands of bridges across the rivers on its way. Some of the views are breathtaking.